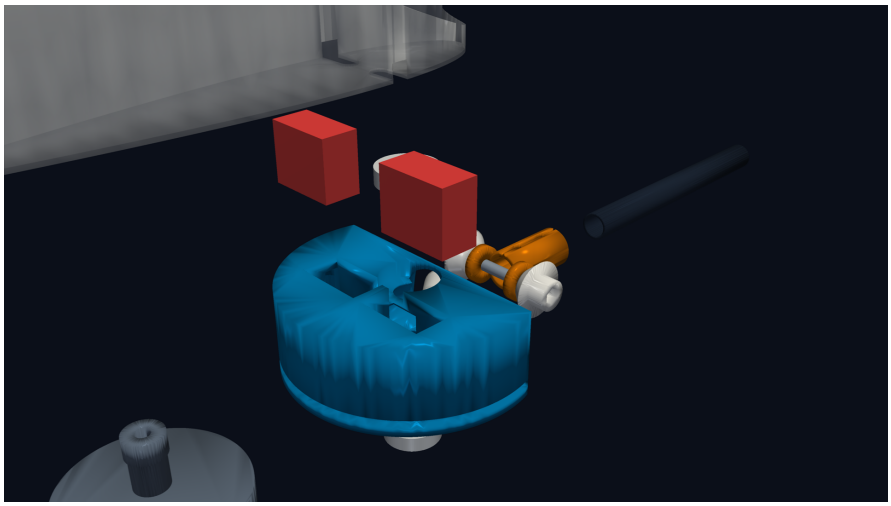
MCO-1D ROBOTIC WING ROOT WRIST ASSEMBLY GUIDE

V10 Stage-Coach Architecture: 10:1 Drivetrain, 25.7g Hollow Spoke Drum, Open Servos & Quick-Swap Bevels



CALLOUT LEGEND	SUBSYSTEM & PART
[1]	NACA 0018 Fairing (0.4mm vase)
[2]	Dual AGFRC B24CLM (7 kg·cm)
[3]	Dual 6800-2RS Bearings (10x19x5)
[4]	Stage-Coach Drum (25.7g)
[5]	Diff Gimbal Yoke & Collar
[6]	3x Quick-Swap Gears (m=1.0)
[7]	Ø4x30mm Steel Dowel Axle
[8]	Ø10mm Carbon Wing Spar
[9]	Stepped Pinion Drum / 5010
[10]	0.75mm Dyneema SK75 Line

1. 3D PRINT MANUFACTURING SPECIFICATIONS & BOM (PER WING ROOT)

v10_stagecoach_drum [STL] [STEP]	PA12-CF / PA6-CF	1x	0.4mm / 0.16mm	7 radiating spokes, 5 rim perimeters, open gates	**25.7 g**	~$1.15
v9_quickswap_bevel_gear [STL] [STEP]	PA-CF or Tough Resin	3x	0.4mm / 0.12mm (35µm)	≥ **5 walls (100% infill)**, flat on back	4.5 g (set)	**$0.20 (set)**
v9_diff_gimbal_yoke [STL] [STEP]	PA12-CF or PET-CF	1x	0.4mm / 0.12mm	6 walls, 100% solid perimeters	8.2 g	~$0.37
v9_topological_fairing [STL] [STEP]	Foaming LW-PLA	1x	0.4mm / 0.20mm	1 perimeter (0.45mm), 0% infill (vase)	5.8 g	~$0.25
v9_stepped_pinion [STL] [STEP]	High-Temp PA-CF / 6061	1x	0.4mm / 0.12mm	100% solid, horizontal orientation	2.1 g	~$0.10

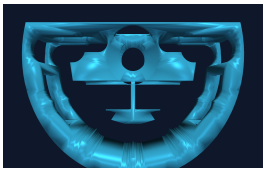
2. HARDWARE CHECKLIST & SLICER QUALITY GATES

OTS Hardware Component	Specification & Grade	Qty	Assembly Function & Torque Spec
Flapping Hub Bearings	6800-2RS (10mm ID x 19mm OD x 5mm W)	2x	Dual radial-thrust couple; light press-fit into drum
Gimbal Cross-Axle	Ø4.0mm x 30.0mm Ground Steel Dowel Pin (h6)	1x	Carries 4g lift bending & bevel gears; zero flex
Actuator Micro-Servos	AGFRC B24CLM (7.0 kg·cm @ 8.4V, Coreless)	2x	Feathering & sweep differential; hex output spline
Capstan Drive Cable	0.75mm Dyneema SK75/SK99 (12-strand braided)	~1m	D:d = 10.0:1 on Ø15.4mm pinion; 180N pre-tension
Fastener Assortment	Grade 12.9 Socket Head Cap Screws	Kit	3x M2x8 (gears), 2x M2.5x12 (clamp), 2x M2.5x6 (cable), 1x M3x5 (pinion)

CRITICAL SLICING MANDATE: Quick-swap bevel gears (v9_quickswap_bevel_gear.stl) **must be sliced with ≥ 5 concentric wall perimeters** (or 100% concentric infill). Slicing with grid/rectilinear infill leaves the 5.05mm hex hub corners unsupported and vulnerable to radial wedging burst under 7.0 kg·cm servo stall torque. Pre-treat gear teeth with dry PTFE aerosol or MoS■ powder before final mesh.

3. STEP-BY-STEP PICTORIAL BENCH ASSEMBLY SEQUENCE

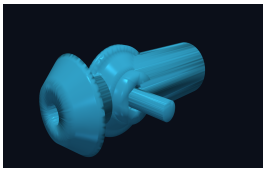
Follow stages 1 through 6 in exact sequence.



STAGE 1: BEARING HUB PRESS-FIT

• Inspect PA12-CF sector drum bearing seats (19.0mm OD).
• Press-fit lower 6800-2RS bearing flush to Z=0.0mm datum.
• Press-fit upper 6800-2RS bearing flush to Z=28.0mm top deck.
• Ensure central Ø10mm wiring pass-through bore is clear.

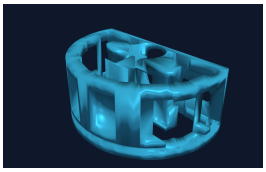
VERIFICATION GATE: Bearings spin freely with zero axial tilt or binding.



STAGE 2: DIFFERENTIAL CARTRIDGE ASSEMBLY

• Slide central spider bevel gear onto yoke aft hex boss.
• Thread M2x8mm axial cap screw into spider gear counterbore (0.35 N·m).
• Insert Ø4.0mm x 30mm ground steel cross-axle through yoke bore.
• Slide left and right input bevel gears onto axle ends.

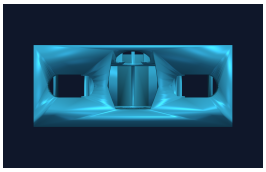
VERIFICATION GATE: Bevel gearset meshes smoothly; backlash between 0.2° and 0.4°.



STAGE 3: SERVO INSTALLATION & WIRING

• Drop dual AGFRC B24CLM micro-servos into isolated bays.
• Verify 1.8mm center web isolates servo cases at X=0.
• Route servo wiring leads down through bottom chimney vents.
• Engage servo output hex splines into left/right input gears.

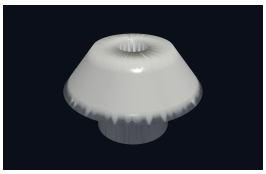
VERIFICATION GATE: Servo case heatsinks align with bottom ram-air scoops.



STAGE 4: WING SPAR CLAMP & TRUMPET FIT

• Insert gimbal yoke assembly through front drum cavity.
• Slide Ø10.0mm carbon spar tube 26mm deep into clamp collar.
• Torque 2x M2.5x12mm pinch bolts to 0.70 N·m (clamp locked).
• Articulate spar: check -15° feathering to +36° downstroke.

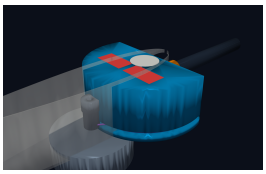
VERIFICATION GATE: No spar contact on trumpet perimeter; ≥ 2.0mm air gap.



STAGE 5: 0.75MM DYNEEMA CAPSTAN RIGGING

• Anchor Dyneema line at terminal -190° with M2.5 clamp washer.
• Wrap line around 200° continuous rim (Z=2.8-5.4mm groove).
• Wrap 3 full figure-8 turns around Ø15.4mm stepped pinion drum.
• Tension line to 180 N; lock M2.5 terminal anchor screw (0.40 N·m).
• Tighten M3x5mm grub screw onto motor shaft D-flat (1.10 N·m).

VERIFICATION GATE: Zero slip at 220 N peak reversal; drum tracks 10:1 ratio.



STAGE 6: NACA 0018 FAIRING & FINAL Q/A

• Slide LW-PLA aerodynamic fairing over assembled wing root.
• Confirm nose aperture centers over carbon spar tube.
• Verify belly NACA scoop connects to servo bottom vents.
• Connect servo harness to flight controller; sweep pitch/roll.

VERIFICATION GATE: Full ±36° pitch and 200° flapping stroke with zero friction.

FIELD PROTOCOL: 90-SECOND QUICK-SWAP BEVEL GEAR REPLACEMENT

Inspection Trigger: If pre-flight bench checks detect gear backlash exceeding **1.0°** or tooth flank wear, initiate cartridge swap:
1. Remove fairing shell (2x M2 nose retaining clips).
2. Unthread axial M2x8mm socket cap screw from worn bevel gear.
3. Slide worn gear off 5.05mm hex drive shaft; slide on fresh 3D-printed replacement gear (v9_quickswap_bevel_gear.stl, pre-treated with dry PTFE).
4. Re-torque M2 screw to 0.35 N·m; verify backlash < 0.3°. Re-install fairing. *Total downtime: < 90 seconds. Cost: $0.07.*